Laboratory Tests for Assessment of Lipoprotein disorders

Lipid Profile



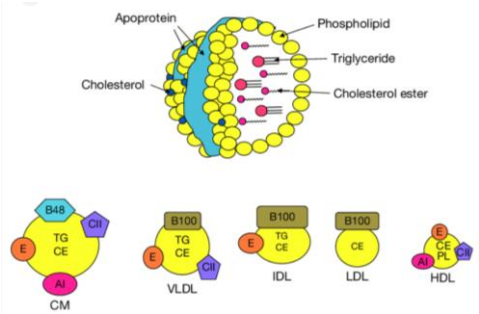
Assoc. Prof. Dr. Wilaiwan Sriwimol

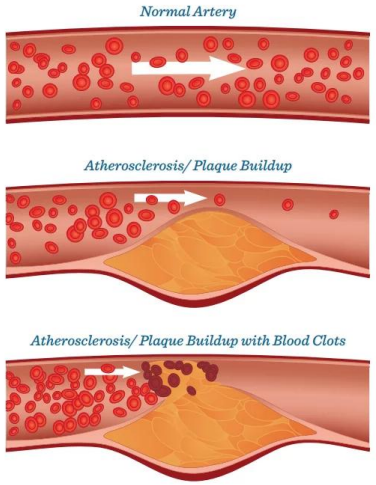
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- **Lecture 12**: Lipoprotein (อ.ลัดดา)
- **SDL 5**: Atherosclerosis (อ.ลัดดา)
- **SDL 6**: Lipid and lipid disorders (อ.สุภมัย)





SDL 7: Laboratory tests for assessment of lipoprotein disorders

วัตถุประสงค์: อธิบายหลักการทดสอบ เลือกใช้และแปลผลการทดสอบทางห้องปฏิบัติการเพื่อประเมินความผิดปกติของไขมันได้

เนื้อหา

- รายชื่อ
- การตรวจวิเคราะห์ Lipid profile: Total cholesterol, Triglyceride, HDL-cholesterol, LDL-cholesterol
- การตรวจวิเคราะห์อื่นๆ เช่น Apolipoprotein A-I, Apolipoprotein B
- ปัจจัยที่มีผลต่อการตรวจวิเคราะห์
- การแปลผล Lipid profile และการใช้ประโยชน์ทางคลินิก

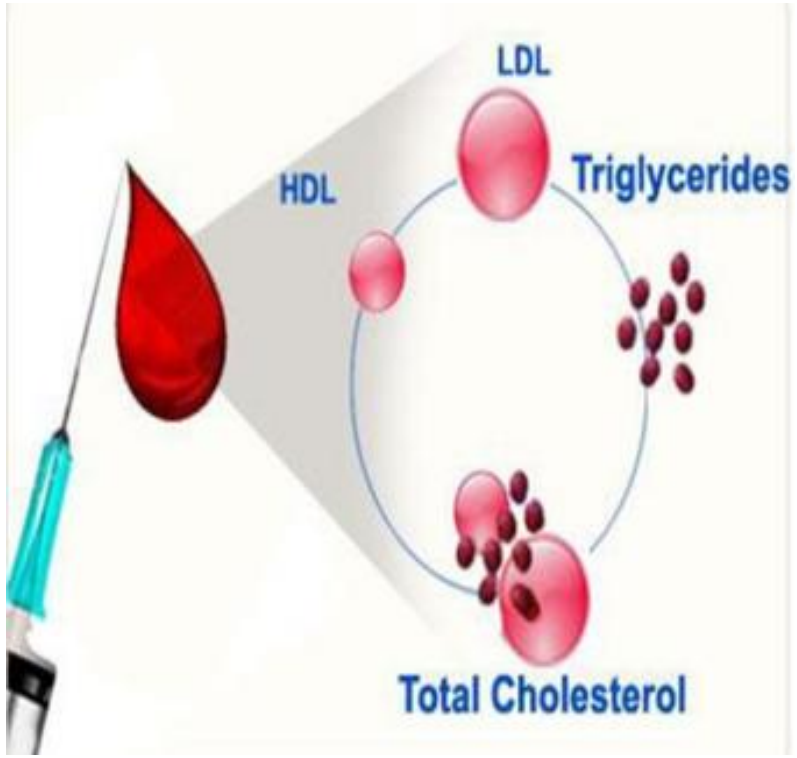
Lipid Profile

Lipid profile/lipid panel is a group of tests including of

- Total cholesterol (TC)
- Triglyceride (TG)
- High density lipoprotein cholesterol (HDL-c)
- Low density lipoprotein cholesterol (LDL-c)

Diagnostic importance of lipid profile

- Diagnosis of dyslipidemia
- To assess the risk of atherosclerosis and coronary artery disease (CAD)



Lipid Profile

Who should be checked ?

- All patients known to have cardiovascular diseases (coronary heart disease, stroke, peripheral vascular disease)
- A family history of premature coronary disease (occurring at age <60 year olds)
- Other major risk factors for CHD: *e.g*. diabetes mellitus, hypertension, smoking, obesity
- Other patients at high risk for hyperlipidemia: *e.g.* thyroid disease, liver disease, renal disease
- *ACC/AHA recommendation*: all adults above the age of 20 if the risk factors are present → check lipid profile at least once every five years
- A children with family history of hereditary hyperlipidemia → start to check lipid profile when the age of 10
- All individuals over 35 year olds→ should check lipid profile at least once every five years

Patient preparation

การเตรียมตัวก่อนเจาะเก็บเลือด

- การส่งตรวจ lipid profile ซึ่งมีรายการตรวจ triglyceride อยู่ด้วยนั้น จะแนะนำให้ผู้ป่วยต้องงดอาหารอย่างน้อย 12-14 ชั่วโมงก่อนเจาะเก็บเลือด โดยสามารถดื่มน้ำเปล่าได้
- ควรรับประทานอาหารที่เคยรับประทานอยู่ตามปกติในระยะเวลา 2-3 สัปดาห์ก่อนมาตรวจ
- ยาบางชนิดอาจมีผลทำให้การตรวจวัดระดับไขมันในเลือดเปลี่ยนแปลงไปจากสภาวะจริงของผู้ป่วย จึงมีข้อแนะนำให้ผู้ป่วยงดยาเหล่านั้นอย่างน้อย 3 วันก่อนเจาะเลือด



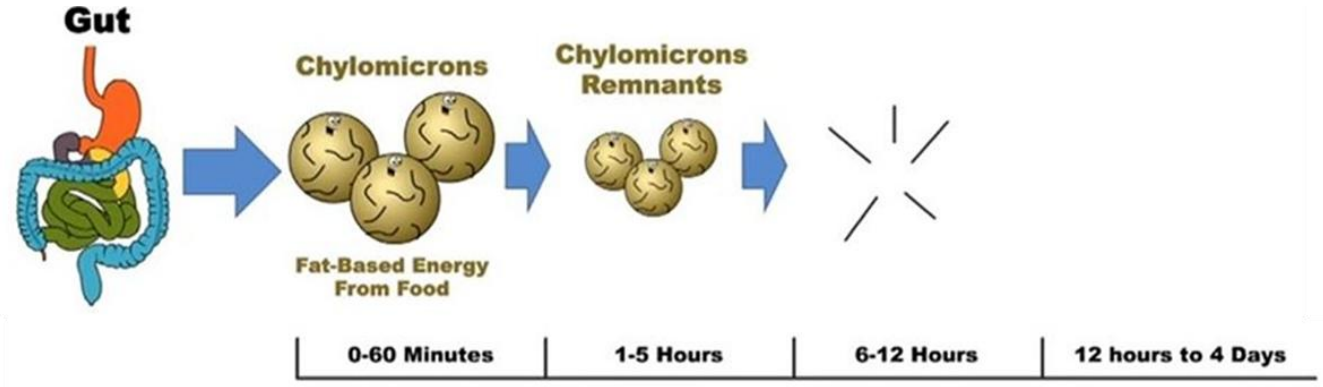
Normal serum/plasma



Lipemic serum/plasma

Patient preparation

Fasting *vs* non-Fasting



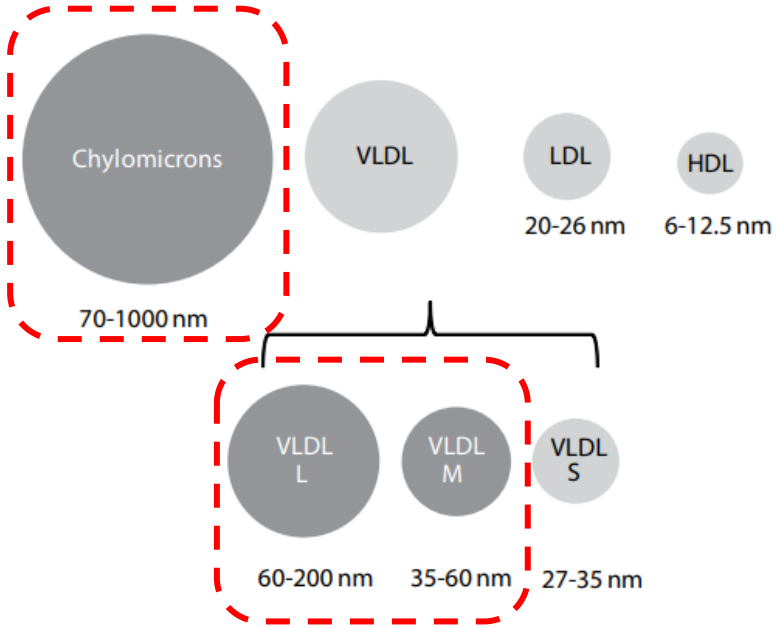


Lipemic serum/plasma



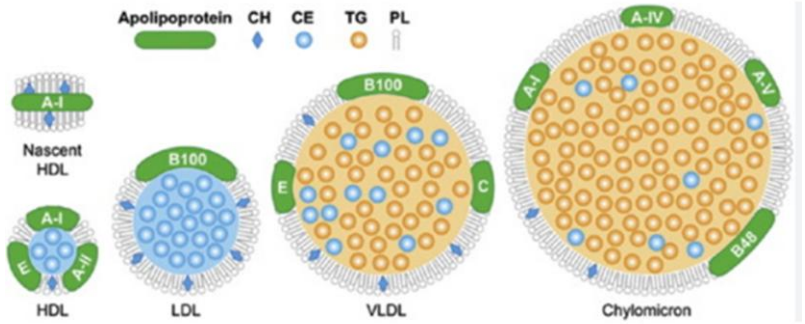
Normal serum/plasma

- **Lipemia** is a turbidity of the sample caused by accumulation of lipoprotein particles.
- As lipoproteins vary in sizes, not all classes contribute equally to the turbidity.



Effects of Lipemia

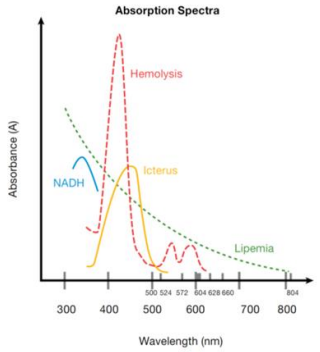
1. False positive of Triglyceride level

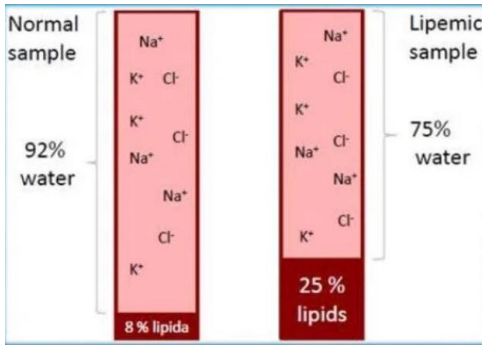


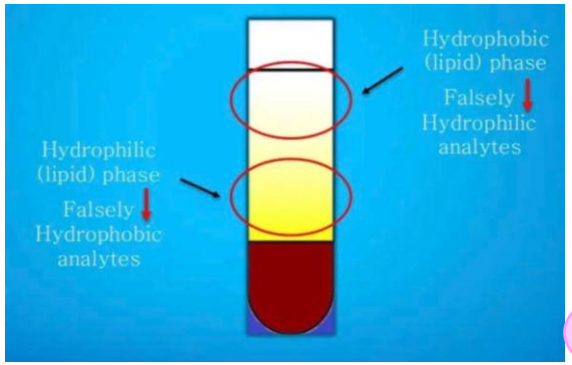
2. To interfere the measurement of other tests

Mechanisms of Interference

1. Light scattering effects may increase absorbance during end point reactions and non-blanked reactions for some analyte
2. Displacement of plasma water: decreases the value of some analyte particularly electrolytes (Na^+, K^+)
3. Hemolysis of RBCs is enhanced in the presence of lipemia
4. Non homogeneity of the sample
5. Physical and chemical interferences

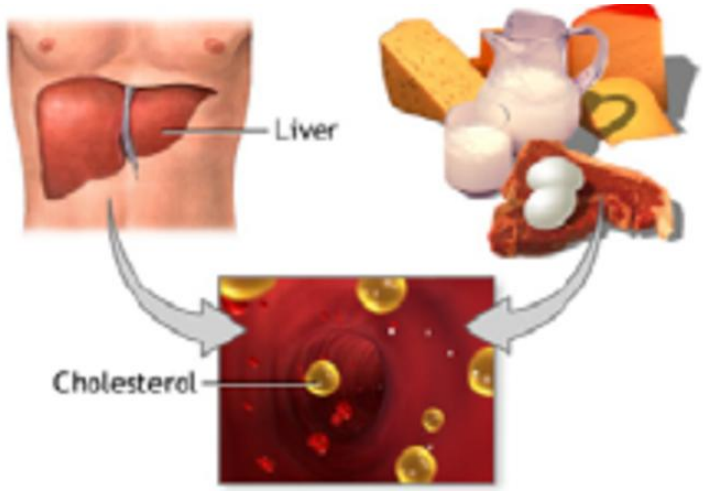




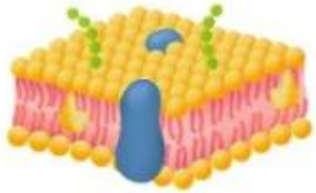


Total Cholesterol (TC)

- Cholesterol is a type of fat, found in your blood.
- Sources of cholesterol:
  - 70% synthesized in the body
  - 30% from foods



- Essential functions of cholesterol
  - Component of cell membrane: integrity, functions, biogenesis, trafficking and signaling



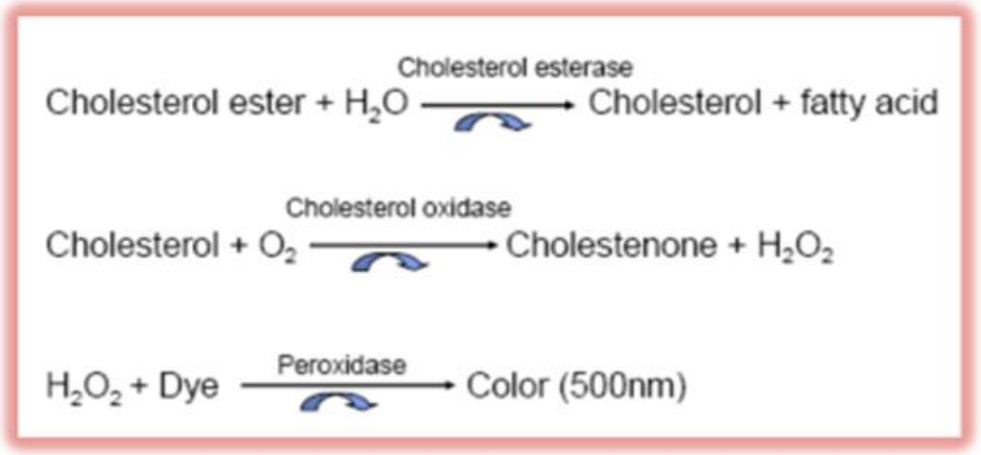
  - Plasma lipoprotein constituent
  - Precursor for steroid hormones
  - Precursor for Vitamin D
  - Precursor for bile acids
  - Insulate nerve fiber

Total Cholesterol (TC)

- This test measures all of cholesterol in all the lipoprotein particles



- Enzymatic colorimetric method



- Interpretation

High	≥ 240 mg/dL
Borderline High	200 – 239 mg/dL
Desirable	< 200 mg/dL

High level → a greater risk for heart disease

Total Cholesterol (TC)

Hypercholesterolemia

- Overweight/Obesity
- Physical inactivity
- High-fat diet
- Diabetes mellitus
- Biliary cirrhosis/obstructive jaundice
- Hypothyroidism
- Nephrotic syndrome
- Genetics: familial combined hyperlipidemia (FH), polygenic hypercholesterolemia
- Certain drugs: anabolic steroids, beta-blockers, epinephrine, oral contraceptives, vitamin D
- Pregnancy

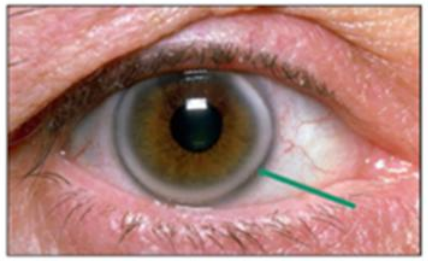
Hypocholesterolemia

- Liver disease
- Hyperthyroidism
- Malabsorption
- Malnutrition
- Celiac disease

etc



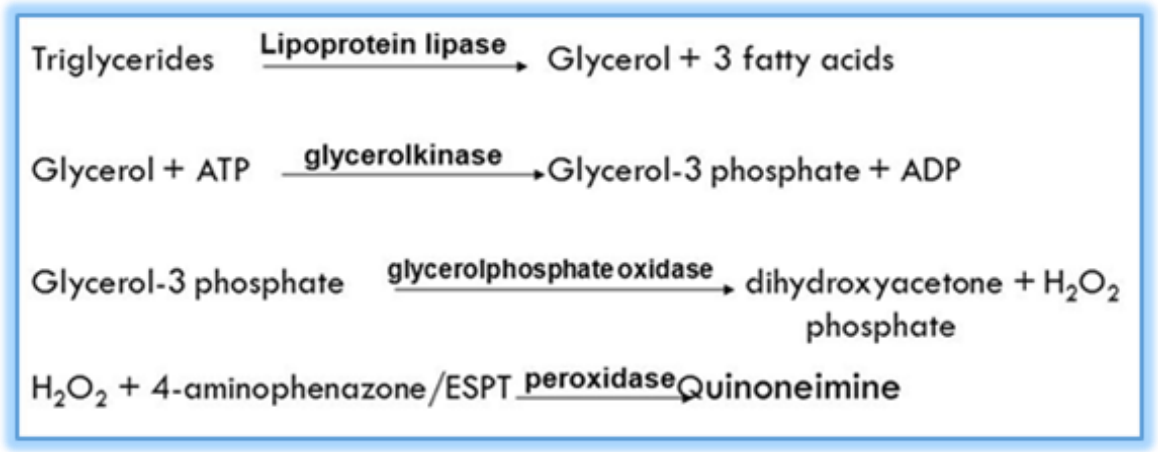




Triglyceride (TG)

- Triglycerides are a type of fat, or lipid, that accounts for the majority of fat in a diet.
- TG is body storage form of fat and energy, found in adipose tissue

➢ Enzymatic colorimetric method



➢ Interpretation

Very High	≥ 500 mg/dL
High	200 – 499 mg/dL
Borderline High	150 – 199 mg/dL
Desirable	< 150 mg/dL

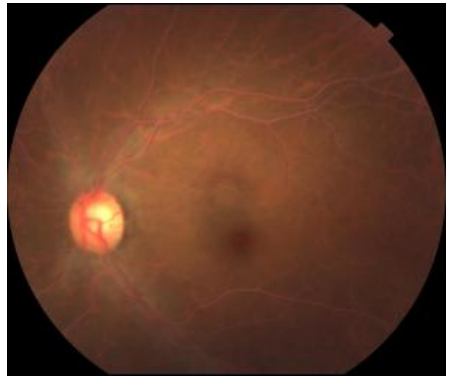
Triglyceride (TG)

Hypertriglyceridemia

- Increases the risk of heart disease and stroke
- Increases the risk for pancreatitis
- Associated with the following clinical findings: *e.g.* eruptive xanthoma, lipemia retinalis, hepatomegaly, splenomegaly

- Overweight/Obesity
- Physical inactivity
- High-carbohydrate diets
- Excess alcohol intake
- Cigarette smoking
- Diabetes mellitus
- Nephrotic syndrome
- Liver disease
- Hypothyroidism
- Certain drugs: *e.g.* anabolic steroids, oral contraceptives, beta-andrenergic blocking agents
- Genetics: hyperlipidemia type I, III, IV, V

etc







<u>Low levels of TG can be present:</u>

- Hyperthyroidism
- Malabsorption syndrome
- Familial hypobetalipoproteinemia

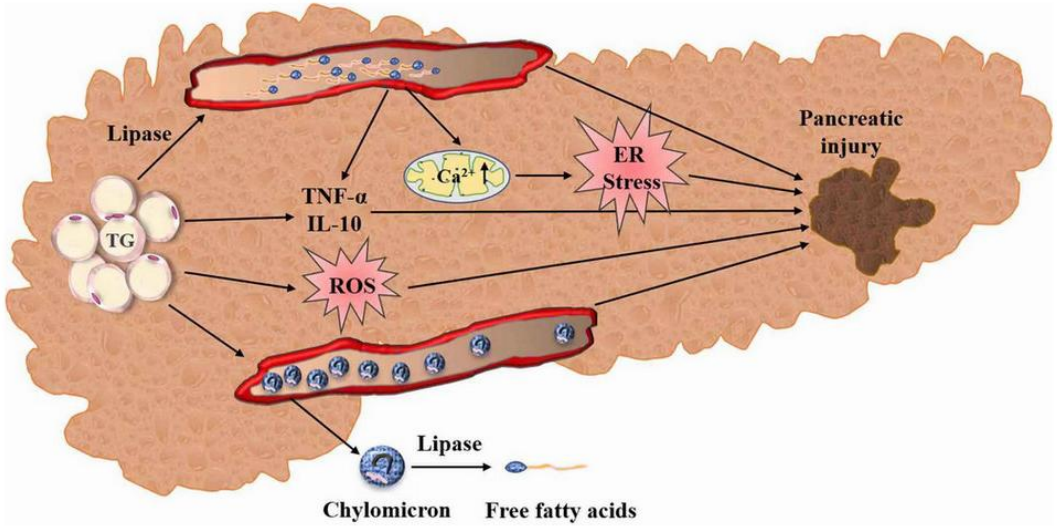
etc

Pancreatitis + Hypertriglyceridemia

Hypertriglyceridemic pancreatitis หรือ Hypertriglyceridemia-induced acute pancreatitis

- **Relation of triglyceride level & risk of pancreatits**
  - Serum/plasma TG is <1,000 mg/dL → hypertriglyceridemic pancreatitis is unlikely
  - Serum/plasma TG above 1,000 mg/dL → the risk increases to 10%
  - Serum/plasma TG above 5,000 mg/dL → the risk increases to above 50%

- **พยาธิกำเนิด → หลักฐานยังไม่ชัดเจน**



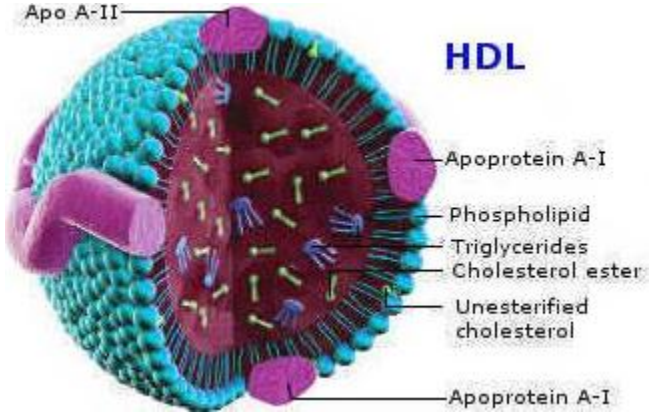
การที่ lipase ย่อยสลาย Triglyceride ที่มีปริมาณมากขึ้น ทำให้เกิดการสะสมของ free fatty acid ในตับอ่อนมากขึ้น เกิดการทำลาย Acinal cell และ Pancreatic capillary ส่งผลให้เกิดการขาดเลือดและสภาวะความเป็นกรดภายในตับอ่อนตามมา

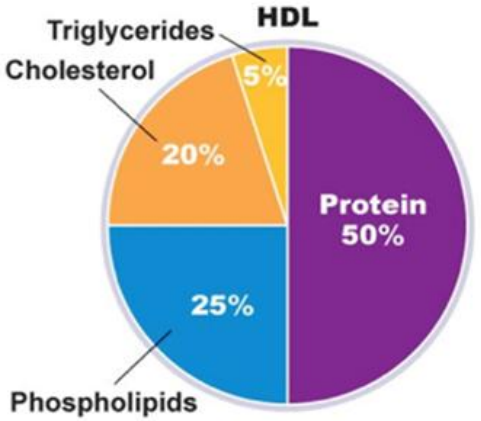
การเกิดภาวะเลือดข้นขึ้น (hyperviscosity) จากการเพิ่มขึ้นของ Chylomicron ของเส้นเลือดฝอยในตับอ่อน ทำให้เกิดการขาดเลือดภายในตับอ่อนขึ้น

Guo YY, *et al.* Hypertriglyceridemia-induced acute pancreatitis: progress on disease mechanism and treatment modalities. Discov Med 2019;27(247):101-9.

Hgh Density Lipoprotein Cholesterol (HDL-c)

- HDL is a fraction of plasma lipoproteins
- Approximately 20-30 % of total cholesterol in serum/plasma
- Major apolipoproteins: apo A-I and Apo A-II
- Good cholesterol; to remove excess cholesterol and carry it to the liver for removal
- High levels linked to a reduced risk of heart and blood vessel diseases. → The higher HDL-c level, the better
- Good cholesterol, cardioprotective



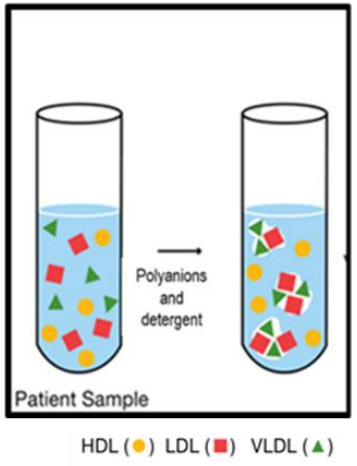


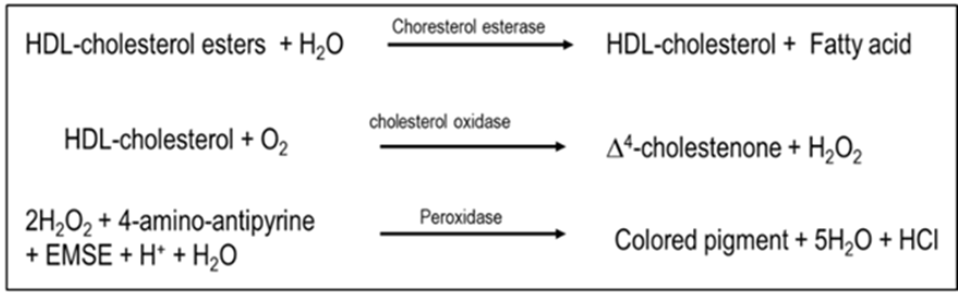


Hgh Density Lipoprotein Cholesterol (HDL-c)

HDL-C => measures the cholesterol in HDL particles

- Homogenous enzymatic colorimetric method





- Interpretation

Low	Men <40 mg/dL Women <50 mg/dL
Normal	Men 40 – 59 mg/dL Women 50 – 59 mg/dL
Ideal	≥ 60 mg/dL

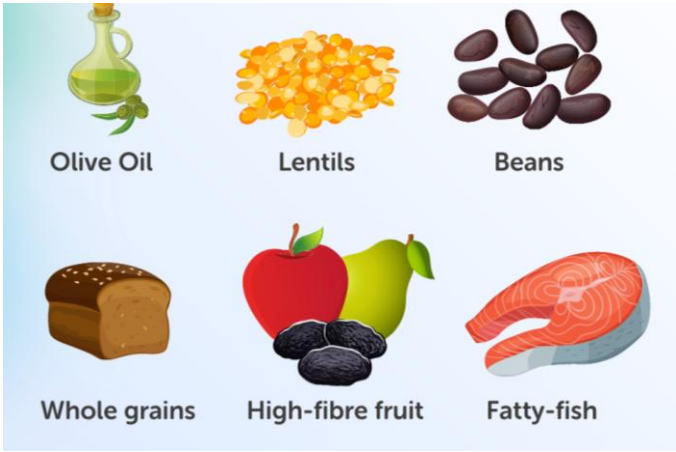
Hgh Density Lipoprotein Cholesterol (HDL-c)

Low HDL-c levels

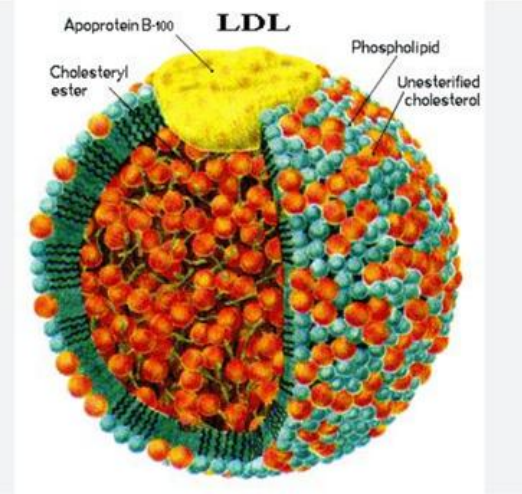
- Overweight/Obesity
- Physical inactivity
- High-carbohydrate diets (>60% of total energy intake)
- Elevated serum/plasma triglycerides
- Cigarette smoking
- Diabetes mellitus
- Genetic factors
- Certain drugs: *e.g.* anabolic steroids, beta-blockings, progestational agents

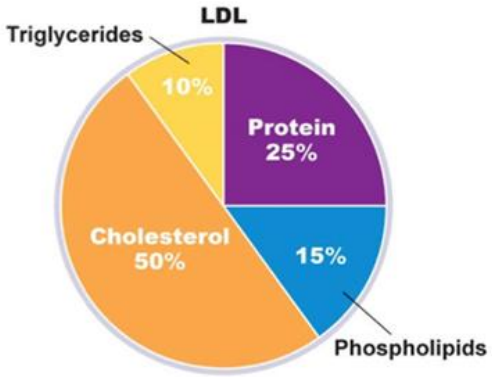
Ways to increase HDL-c level

- Exercise regularly
- Consuming olive oils and other health fats
- Low-carbohydrate diet
- Non-smoking
- Low weight

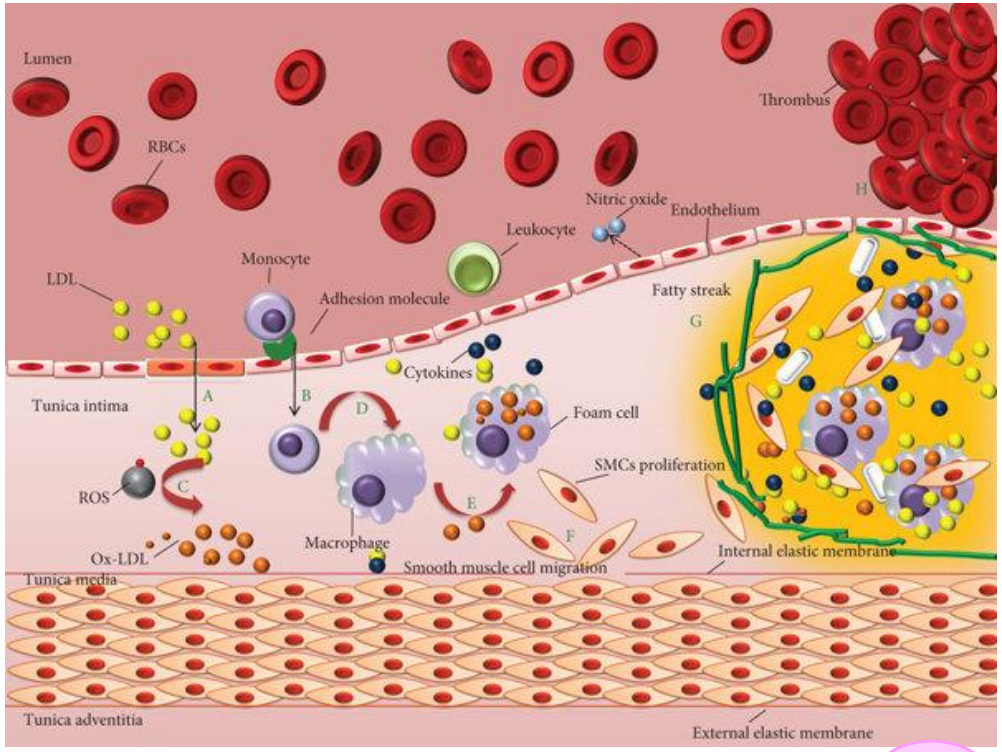


Low Density Lipoprotein Cholesterol (LDL-c)





- LDL is a fraction of plasma lipoproteins
- Approximately 60-70% of total cholesterol in serum/plasma
- Major apolipoproteins: apo B-100 (apo B)
- Bad cholesterol; major atherogenic lipoprotein
- High LDL-c → high risks of atherosclerosis & CHD
- Primary target of cholesterol-lowering therapy



Low Density Lipoprotein Cholesterol (LDL-c)

LDL-C => measures the cholesterol in LDL particles

- *Indirect LDL-c หรือ calculated LDL-c (cLDL-c)*
- *Direct LDL-c (dLDL-c)*

Calculated LDL-c (cLDL-c)

(Indirect LDL-c)

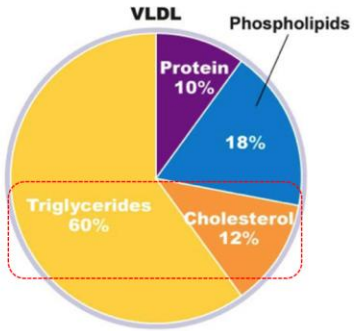


- Calculate from the ***Friedewald formula/equation***

Total cholesterol = VLDL-c + LDL-c + HDL-c

LDL-c = Total cholesterol - VLDL-c - HDL-c

LDL-c = Total cholesterol – TG/5 - HDL-c



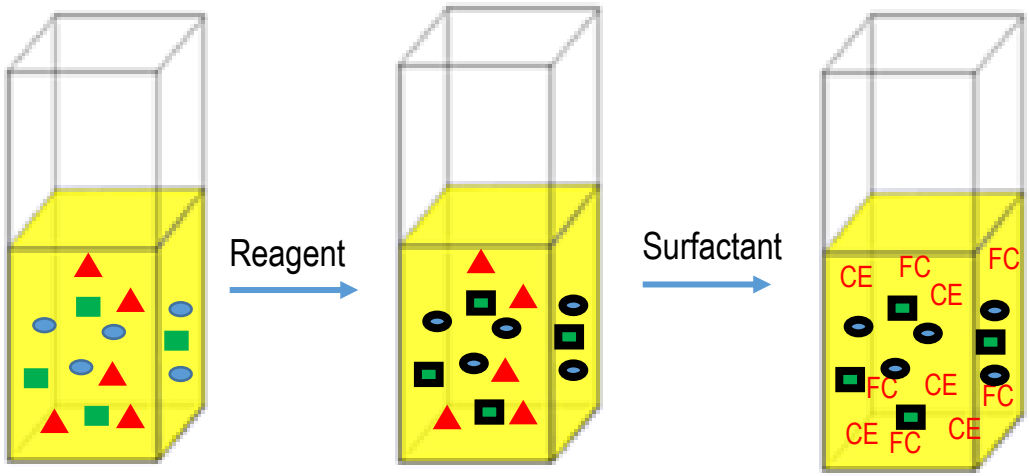
Limitations of the Friedewald formula

- When chylomicrons are present
- When plasma triglyceride concentration exceeds 400 mg/dL
- In patients with type III hyperlipoproteinemia

Low Density Lipoprotein Cholesterol (LDL-c)

Direct LDL-c (dLDL-c)

Homogeneous enzymatic colorimetric assay



LDL-cholesterol esters + H_2O —(detergent, Cholesterol esterase)→ cholesterol + free fatty acids

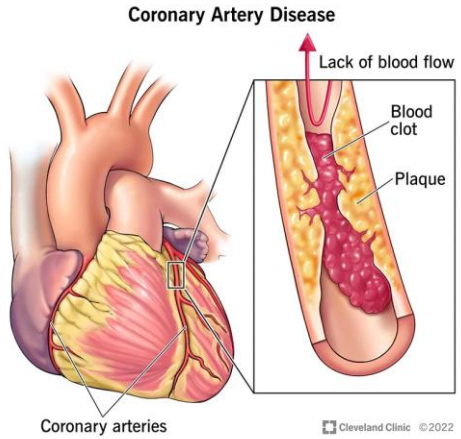
LDL-cholesterol + O_2 —(Cholesterol oxidase)→ Δ^4-cholestenone + H_2O_2

H_2O_2 + 4-amnoantipyrine + HSDA + H_2O + H^+ —(peroxidase)→ red purple pigment +$5H_2O$

➢ Interpretation

Very high	≥ 190 mg/dL
High	160 – 189 mg/dL
Borderline high	130 – 159 mg/dL
Fairly good	100 – 129 mg/dL
Normal	< 100 mg/dL

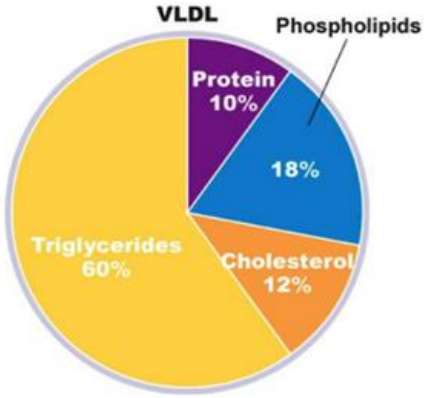
Interpretation of Lipid Profile → CHD



	Ideal (normal)	Moderate risk	High risk
Total cholesterol to HDL-c ratio			
For men	<3.5	3.5 – 5.0	>5.0
For women	<3.0	3.0 – 4.4	>4.4
LDL-c to HDL-c ratio	<2.5	2.5 – 3.3	>3.3
Triglyceride to HDL-c ratio	<3.0	3.1 – 3.8	>3.8

Very Low Density Lipoprotein Cholesterol (VLDL-c)

- Calculated from “Triglyceride/5”



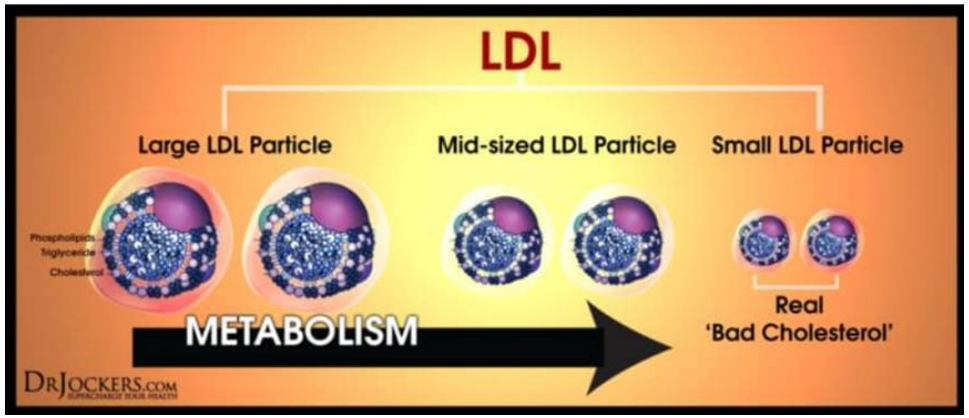
Non-HDL Cholesterol

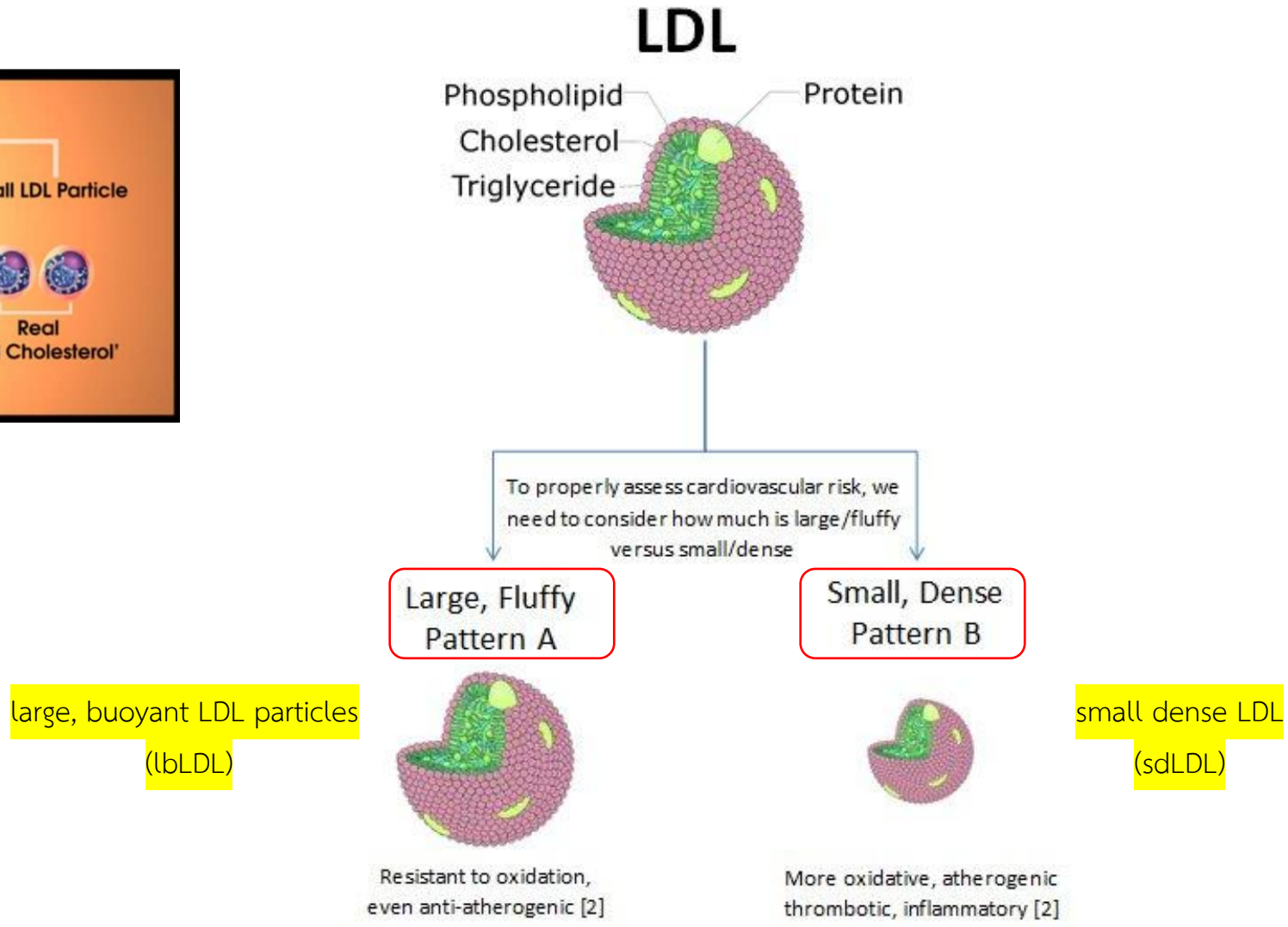
- Since VLDL-c is highly correlated with atherogenic remnant lipoproteins, it can reasonably be combined with LDL-c to enhance risk prediction when serum triglycerides are high.

Non-HDL cholesterol = VLDL-c + LDL-c

Calculated from → Non-HDL cholesterol = Total cholesterol - HDL-c

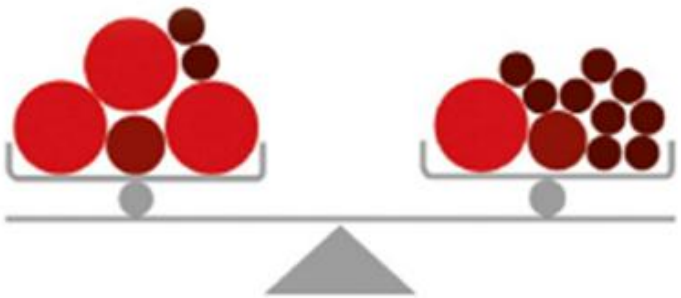
Small Dense LDL (sdLDL)





Small Dense LDL (sdLDL)

LDL-c = 110 mg/dL



Lower risk	Higher risk
sdLDL-c 20 mg/dL	sdLDL-c 70 mg/dL



- Rapid entry into arterial wall
- Less able to bind to LDL receptors
- More susceptible to oxidative damage → oxidized LDL
- More able to bind to scanvanger receptors on macrophage → foam cell
- Poor blood vessel relaxation

Small Dense LDL (sdLDL)

- sdLDL is not routinely measured
- Methods:
  - Density gradient ultracentrifugation (DGUC)
  - Gradient gel electrophoresis (GGE)
  - Nuclear magnetic resonance
  >> Homogenous enzymatic colorimetric

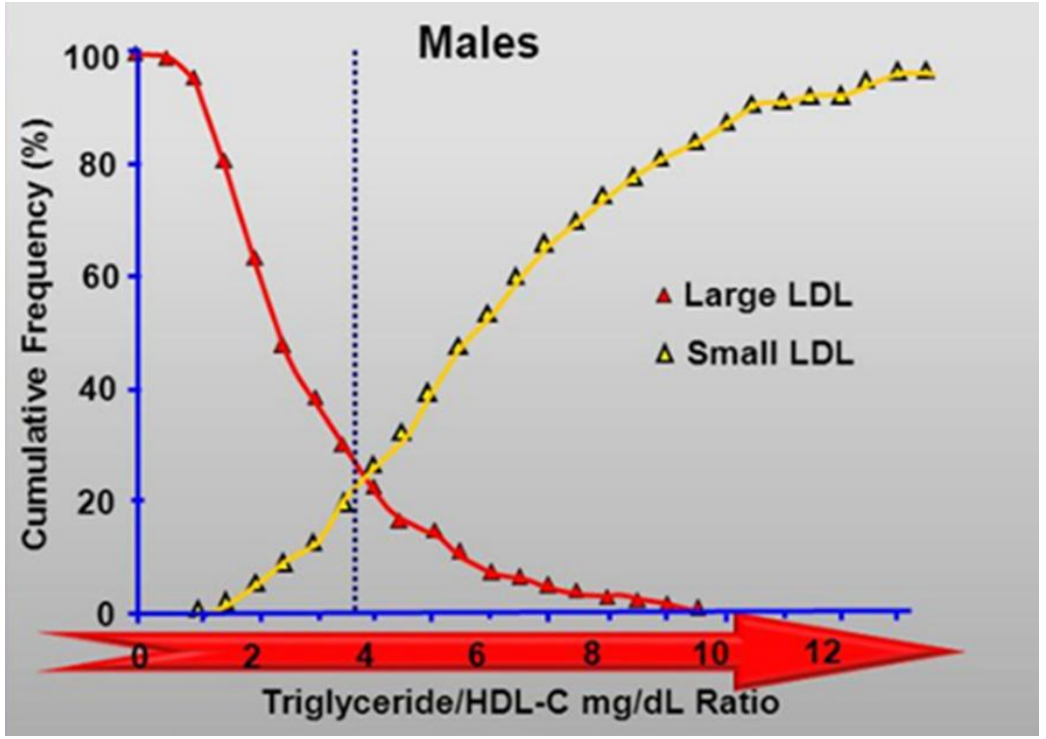
➢ Calculated from

$$\text{sdLDL-C (mg/dL)} = 0.580\ (\text{non-HDL-C}) + 0.407\ (\text{dLDL-C}) - 0.719\ (\text{cLDL-C}) - 12.05$$

Srisawasdi P, et al (2011)

dLDL-c = direct LDL-c
cLDL-c = calculated LDL-c

Relationship of sdLDL to TG/HDL-c Ratio



Hanak, V. et al. Am J Cardiol 2004;94:219-22.

Other Laboratory Tests





- **Apolipoprotein A-1 (Apo A-1) and apolipoprotein B (Apo B)** are used alongside other routine lipid tests to help determine an individual's risk of developing CVD

- Apo B concentrations may also be used to monitor the effectiveness of lipid treatment as an alternative to non-HDL-C
- In rare cases, an Apo B test may be requested to help diagnose a genetic problem that causes over- or under-production of Apo B.
- Apo A-1 levels may be used to help diagnose rare conditions that cause Apo A-I deficiencies, such as Tangier and fish eye disease.

POST-TEST EXAMINATION





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Thank you
for your
attention